Faculty of Engineering
The University of Hong Kong

CIVL2108 PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

Course Outline

Course code:	CIVL2108
Course title:	Principles of Civil Engineering Management
Offered by (department):	Department of Civil Engineering
Level:	2
Credit units:	6
Pre-requisite course code:	Nil
Co-requisite course code:	Nil
Learning hours:	Lecture (26 hours) Tutorial (13 hours)
Student quota:	120
Course type:	Core
Offer in academic year:	1st and 2nd Semester
Assessment (%):	Written examination (50%) Continuous assessment (50%) Practical work (0%)

This document contains information for the course CIVL2108: Principles of Civil Engineering Management.

Its intention is for students undertaking the course to be well informed in terms of its learning expectations and also to help make their learning journey an enjoyable one.

Information presented in this document was correct at the time of printing.

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Course Description and Aims

This is a certified Communication-intensive (CI) Course which meets all of the requirements endorsed by HKU's Senate, including
• the teaching and assessment of visual and written communication 'literacies'; and
• at least 40% of the course grade assigned to communication-rich assessment tasks.

Civil engineering projects are characterised by their uniqueness, complexity and uncertainty, and these have posed immense challenges to our industry. To satisfy the client and project requirements, a good management skill and knowledge is of paramount importance. While engineers play a key role in relevant government departments, client organisations, design offices and contracting firms, they have the responsibility of improving the efficiency, safety and quality of civil engineering projects and maximising the chance of project success; and discharging their duties ethically. Therefore, the aims of this course are to introduce the basic concepts of various aspects of management and to explain how to apply these management principles to plan, organise and control a civil engineering project.

By the end of this course, students should have a good understanding on the operation of the construction industry in particular the civil engineering sector. They should also know what are the roles and responsibilities of engineers at both the pre-contract and post-contract stages of a civil engineering project. Particular emphasis will be placed on the time and cost management techniques, so that students will be able to produce a simple construction programme and a cost estimate at different stages of a project. Besides, students are expected to know how to select a procurement system, contractual method and tendering approach under various circumstances. They should also have the basic knowledge on measurement, tender documents, contract administration, professional ethics as well as the safety and quality management approaches.

Teaching Staff

Contact details for the course co-ordinator and course teachers are shown below:

Course Co-ordinator	
Name (Department: Specialty)	**Contact Details**
Prof. W.J. Cao	E-mail: wjcao@hku.hk Phone: 3917 1977 Office: HW 6-19A, Haking Wong Building
Course Teachers	
Name (Department: Specialty)	**Contact Details**
Prof. W.J. Cao	E-mail: wjcao@hku.hk Phone: 3917 1977 Office: HW 6-19A, Haking Wong Building
Prof. S. Zhan	E-mail: szhan@hku.hk Phone: 3917 1976 Office: HW 5-26, Haking Wong Building

Learning Outcomes and Alignment with HKIE Abilities for Engineering Graduates

By the end of this course students should be able to demonstrate a threshold level of mastery of the following learning outcomes. The appropriate HKIE abilities associated with each of the learning outcomes are also presented.

Course Learning Outcomes	**Relevant HKIE Abilities for Engineering Graduates (Equivalent to Programme Learning Outcomes)**
1. Distinguish various construction stages recognise the roles and responsibilities of professional engineers; select suitable organisation structure for a civil engineering project.	PO(h) function effectively as an individual, and in different roles of diverse and inclusive teams and in various settings as needed; PO(g) apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws with the understanding of the need for diversity and inclusion ; PO(f) analyse and evaluate the impact of engineering solutions in a global, economic, environmental and societal context, especially the importance

Course Learning Outcomes	Relevant HKIE Abilities for Engineering Graduates (Equivalent to Programme Learning Outcomes)
	of health, safety, legal, and sustainable development* considerations to both workers and the general public when solving complex engineering problems.
2. Evaluate different procurement methods, contractual arrangement approaches and tendering alternatives;	PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex 10 engineering problems; PO(g) apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws with the understanding of the need for diversity and inclusion.
3. Evaluate the financial feasibility of a construction scheme; employ the fundamental principle of estimation to derive a unit rate.	PO(c) design creative and sustainable solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate economic, environmental, social, political, ethical, cultural, health and safety, whole-life cost, net zero carbon, manufacturability considerations as required; PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex 10 engineering problems.
4. Discuss the stages involved in planning, organising and controlling the time of a construction project; employ critical path method to prepare a construction programme; apply Microsoft Project to create a gantt chart and level the resources.	PO(c) design creative and sustainable solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate economic, environmental, social, political, ethical, cultural, health and safety, whole-life cost, net zero carbon, manufacturability considerations as required; PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex 10 engineering problems .
5. Recognise the rights and responsibilities of different parties in a contract; employ the contractual knowledge to administer a civil engineering project	PO(h) function effectively as an individual, and in different roles of diverse and inclusive teams and in various settings as needed PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex 10 engineering problems PO(g) apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws with the understanding of the need for diversity and inclusion
6. Understand the roles and responsibility of project managers	PO(c) design creative and sustainable solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate economic, environmental, social, political, ethical, cultural, health and safety, whole-life cost, net zero carbon, manufacturability considerations as required; PO(h) function effectively as an individual, and in different roles of diverse and inclusive teams and in various settings as needed; PO(f) analyse and evaluate the impact of engineering solutions in a global, economic, environmental and societal context, especially the importance of health, safety, legal, and sustainable development considerations to both workers and the general public when solving complex engineering problems.
7. Discuss the concept of circular economy and its application in civil engineering	PO(c) design creative and sustainable solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate economic, environmental, social, political, ethical, cultural, health and safety, whole-life cost, net zero carbon, manufacturability considerations as required; PO(h) function effectively as an individual, and in different roles of diverse and inclusive teams and in various settings as needed

Course Teaching and Learning Activities and Alignment with Learning Outcomes

Apart from face-to-face lectures, tutorials will be arranged to allow students work through the examples and exercises. The Faculty of Engineering has acquired a computerised programming tool - Microsoft Project, and students will have an opportunity to practice using this software during the tutorial sessions. Students will also have to carry out their own research to complete assignment(s) relevant to the identified learning outcome(s).

The course contents and its alignment with the course learning outcomes and assessment tasks are described below.

Course Content	Responsible Lecturer	Course Learning Outcomes
Construction industry overview	Prof. W.J. Cao	1, 5
Project life cycle and roles of and responsibilities of project participants	Prof. W.J. Cao	1, 5, 6
Circular economy and its application in civil engineering	Prof. W.J. Cao	7
Management theory and practice	Prof. S. Zhan	1, 3, 4, 6
Procurement methods	Prof. S. Zhan	2, 5
Contractual arrangement and tendering approaches	Prof. S. Zhan	2, 5
Quality management	Prof. S. Zhan	2, 5
Labour and material management	Prof. S. Zhan	5, 6
Professional ethics	Prof. W.J. Cao	2, 6
Construction planning	Prof. W.J. Cao	4
Financial feasibility study	Prof. W.J. Cao	3
Cost estimation	Prof. W.J. Cao	3

Guidelines on Grading and Standards of Assessment

HKU has adopted standards-based assessment. The general course grade descriptors aim to provide descriptions on the normal expectations of student achieving particular grades.

Course Grade	Description
A	Exceptionally good performance demonstrating a superior understanding of the subject matter, a foundation of extensive knowledge, a skill use of concepts and/or materials, and an ability to analyse and evaluate problems.
B	Good performance demonstrating capacity to use the appropriate concepts, a good understanding of the subject matter, and ability to handle the problems and materials encountered in the course.
C	Adequate performance demonstrating an adequate understanding of the subject matter, ability to handle simple problems, and adequate preparation for moving on to more advanced work in the field.
D	Minimally acceptable performance demonstrating at least partial familiarity with the subject matter and some capacity to deal with relatively simple problems, but also demonstrating deficiencies serious enough to make it inadvisable to proceed further in the field without additional work.
F	Unacceptable performance demonstrating unfamiliarity with the subject matter, and lack of capacity to deal with relatively simple problems, and also demonstrating deficiencies serious enough to make it advisable to retake the course.

Course Assessment Tasks and Alignment with Learning Outcomes

Assessment tasks in this course are described below, which includes weighting, assessment type, and alignment with course learning outcomes.

Weighting of continuous assessment and written examination

Assessment Type	Percentage of Total Assessment (%)	Description
Continuous Assessment	50%	1. Calculations on financial feasibility 2. Construction programming and planning 3. Mock tendering report and presentation
Written Examination	50%	3-hour written examination
Practical Work	0%	N.A.

Course and Assessment Policy

Late Submission / Missed quiz
The Department of Civil Engineering's policy on the submission of late assignments can be found on the Department of Civil Engineering's website.

Academic Conduct
The University Regulations on academic misconduct will be strictly enforced. Students are strongly recommended to refer to http://www.hku.hk/student/plagiarism/ for further details. In addition, students should familiarise themselves with http://lib.hku.hk/turnitin/.

Means / Processes for Student Feedback on Course

SFTL
Students are asked to complete this evaluation (Student Feedback on Teaching and Learning) of their learning experiences at the conclusion of the course. Questionnaire items relate to the overall evaluation of the course as well as an evaluation of teaching.

Other means of student feedback
Students' feedback may be obtained through face-to-face interactions with students during and after the classes, e.g. by finding out whether the pace is too fast and do they have any problem understanding the teaching and learning materials.

Date: Jul 2026 Completed by: Prof. W.J. Cao

Learning Resources

Required text / References
A list of references relevant to the topic taught will be provided in the handout.

Course website
Departmental intranet and email may be used from time-to-time to announce important issues and students are therefore advised to consult these electronic platforms on a regular basis to obtain up-to-date information for this course.